

Advanced Computer Architecture Lab 1: MIPS-R10000

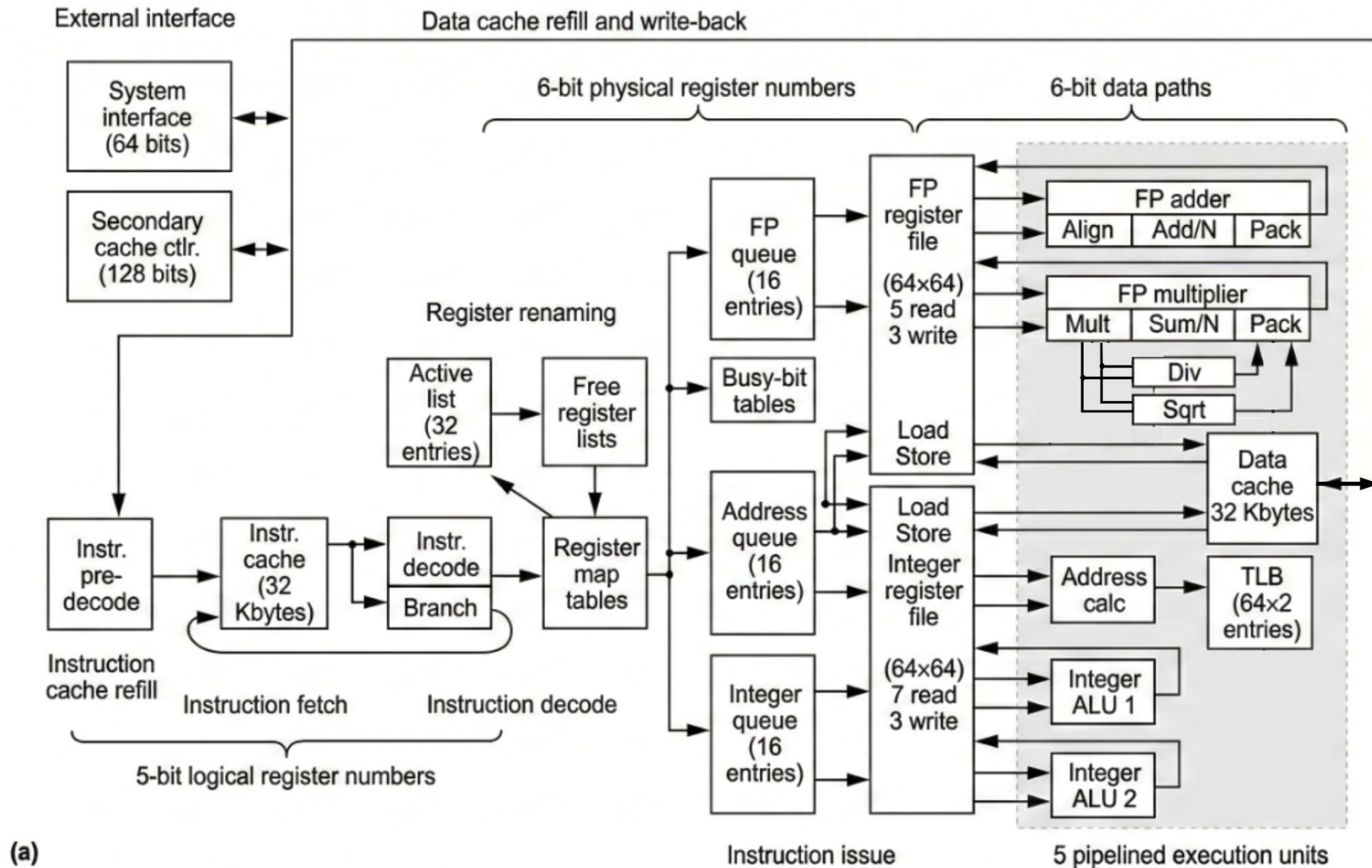
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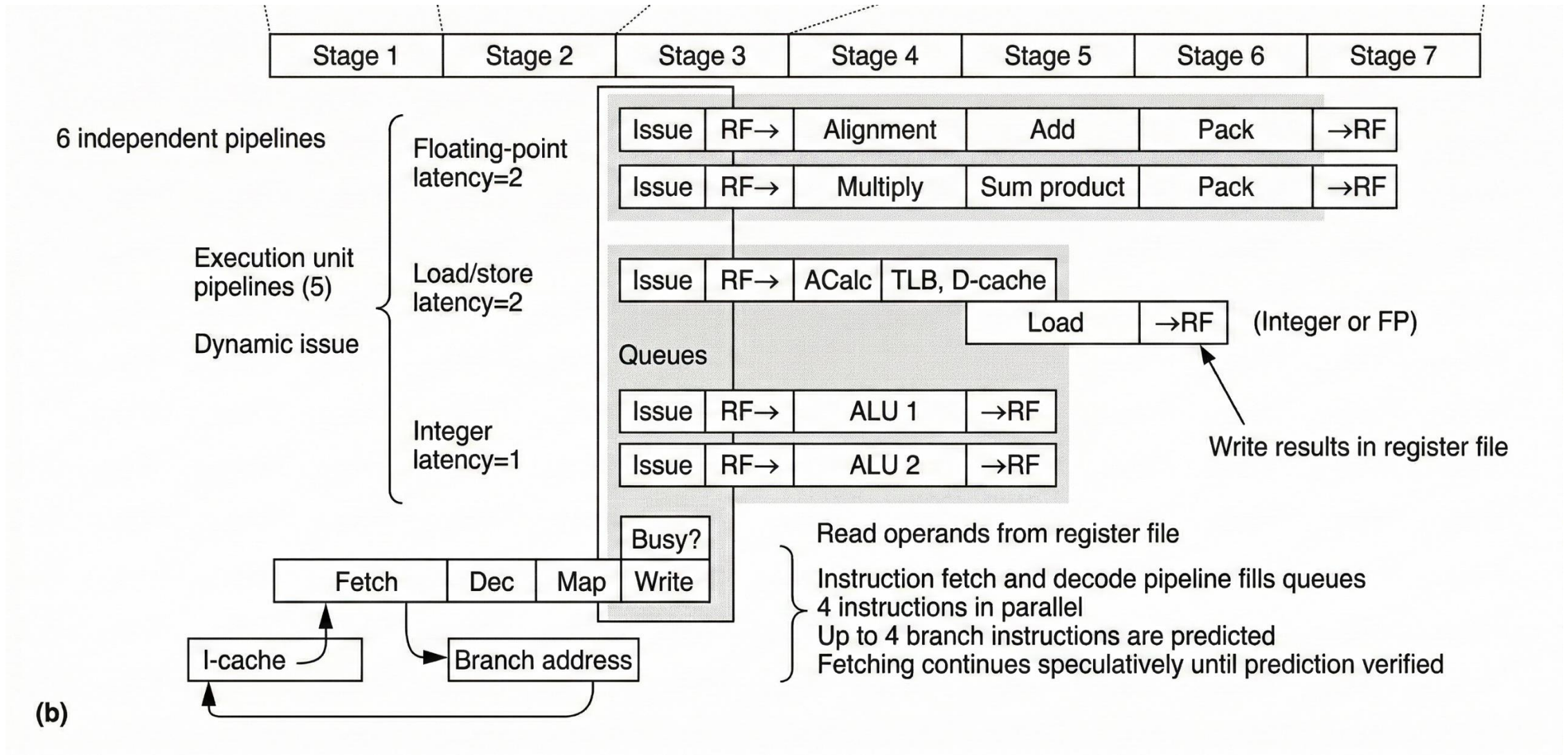
Outline

- Overview of the hand simulation of execution on the R10000 processor
 - Detailed review of the first 8 instructions
- Announcement of Exercise Set 1
- Next session: Q&A Homework, Lab, Exercise Set

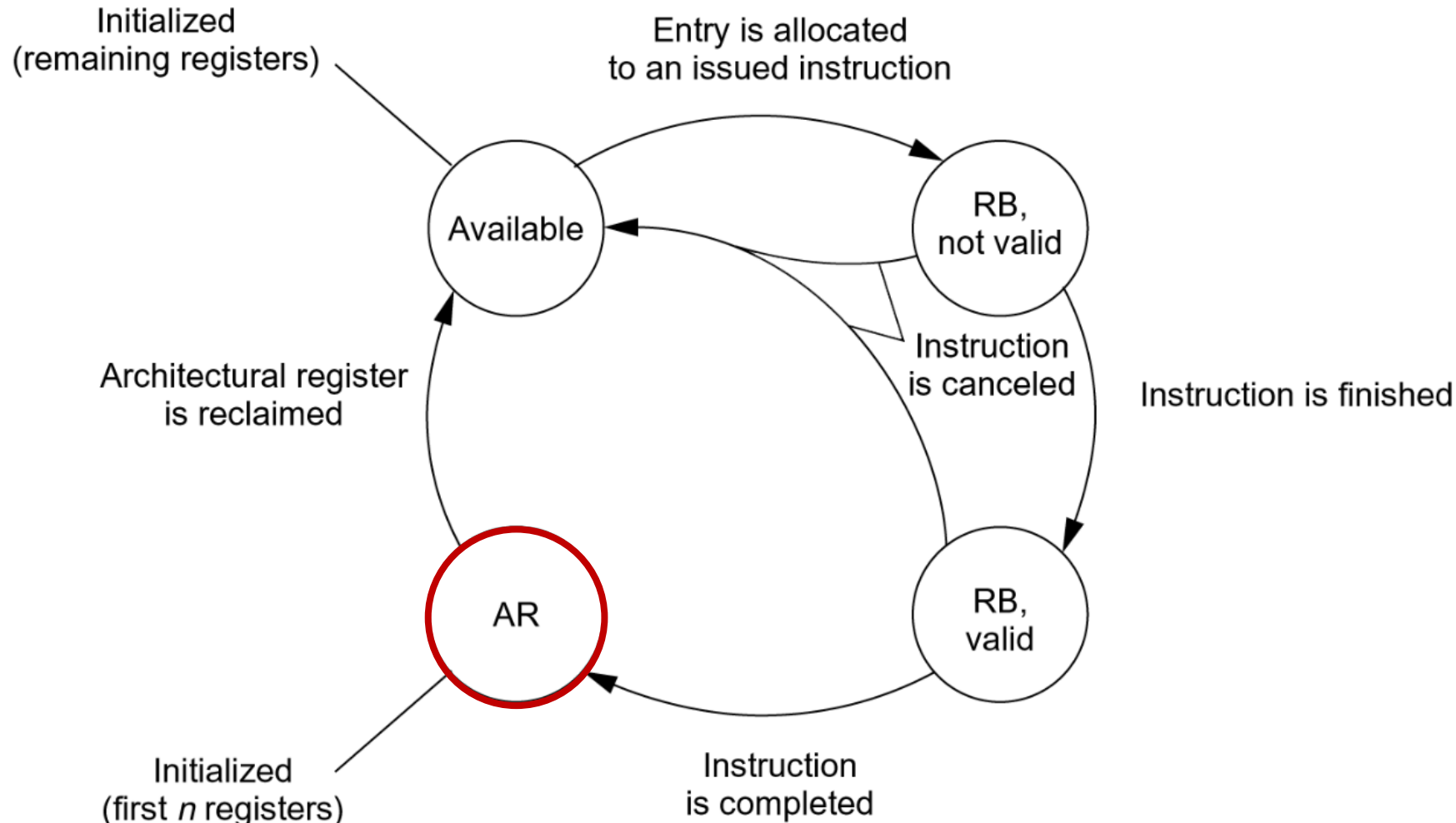
R10000 – Paper Overview



R10000 – Paper Overview



Excel Simulation – Initialization



Physical Registers
x00 to x1f

1 FSM per physical register

Excel Simulation – Cycle 0 (Fetch)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Focus on instructions x01 to x04

- Simplest case: no dependencies
- With full review for x01 and x02

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
1 2 3 4						

1. Fetch first 4 instructions

Excel Simulation – Cycle 1 (Decode)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)

#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Free List								
FIFO1	x20	x24	x28	x2C	x30	x34	x38	x3C
FIFO2	x21	x25	x29	x2D	x31	x35	x39	x3D
FIFO3	x22	x26	x2A	x2E	x32	x36	x3A	x3E
FIFO4	x23	x27	x2B	x2F	x33	x37	x3B	x3F

FIFO Head

FIFO Tail

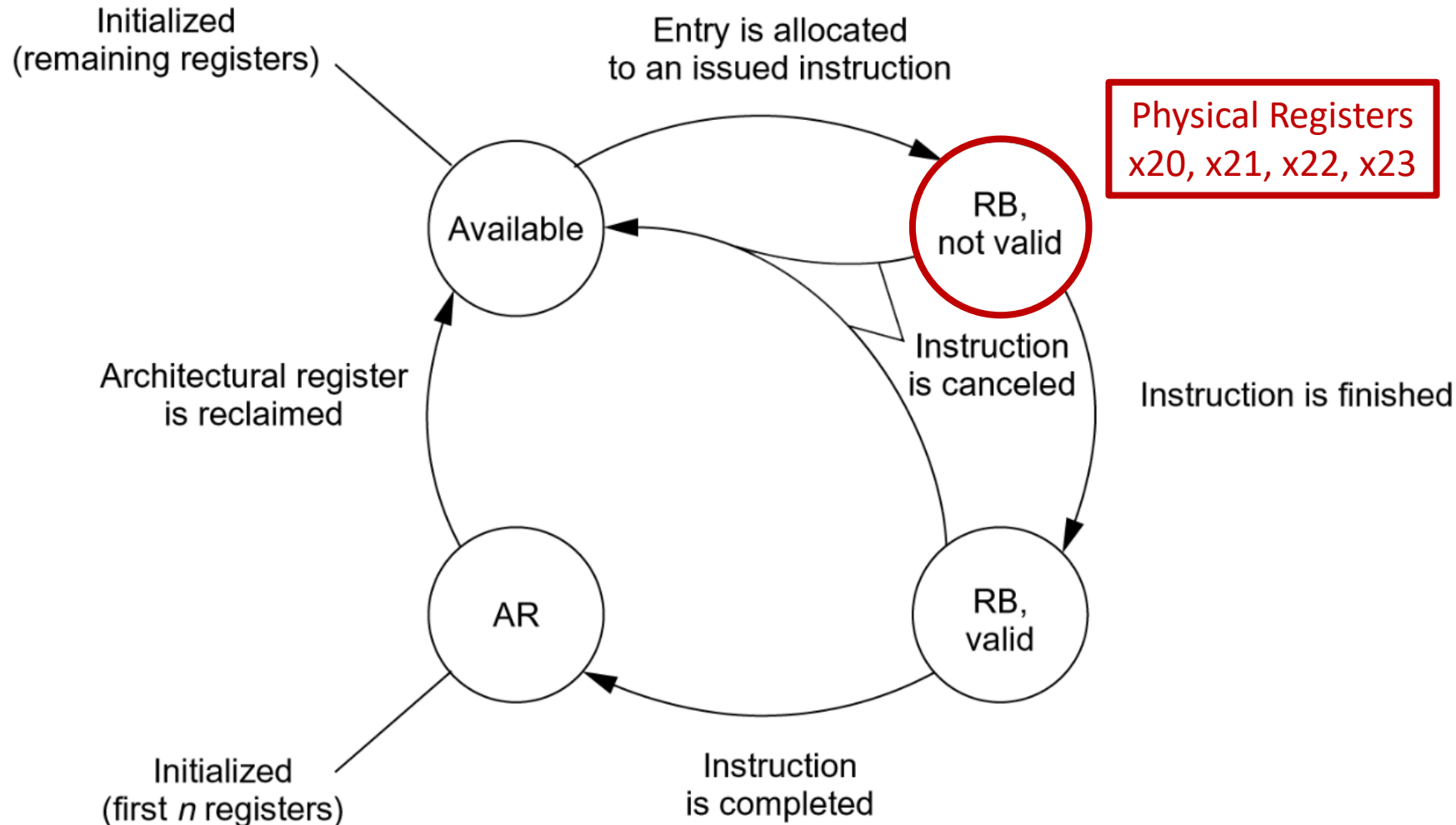
FP Register Map Table	
Logical	Map
x00	x20
x01	x21
x02	x22
x03	x23
⋮	

Busy Bit Table	
⋮	
x20	1
x21	1
x22	1
x23	1
⋮	

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
5 6 7 8	1 2 3 4					

- Free 4 physical registers from the Free List and map them in the Register Map Table
- Mark them as busy in the Busy Bit Table

Excel Simulation – Cycle 1 (Decode)



Physical Registers
x20, x21, x22, x23

Busy Bit Table	
⋮	
x20	1
x21	1
x22	1
x23	1
⋮	

1 FSM per physical register

Excel Simulation – Cycle 1 (Decode)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Active List 1			
Address	OldDest	LogDest	Done
x00	x00	x00	0
:	:		

Active List 3			
Address	OldDest	LogDest	Done
x10	x02	x02	0
:	:		

Active List 2			
Address	OldDest	LogDest	Done
x08	x01	x01	0
:	:		

Active List 4			
Address	OldDest	LogDest	Done
x18	x03	x03	0
:	:		

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
5 6 7 8	1 2 3 4					

3. Allocate entries in the Active Lists

- Address == TAG: On-going instruction identifier
- OldDest: previous physical register assigned to the current logical register (LogDest)

Excel Simulation – Cycle 2 (Issue)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Address Queue			
Tag	Rdy	OpA	Dest
x00	1	I10x01	x20
x08	1	I10x01	x21
x10	1	I10x01	x22
x18	1	I10x01	x23

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
9 A B C	5 6 7 8	1 2 3 4				

1. Push instructions to the Address Queue
 - Tag: Reference to Active List entry
 - Rdy: Is the operand ready? (second operand is constant)
2. Read operand from Register File if ready

Excel Simulation – Cycle 3 (Execution)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

1. Instruction x01 is loaded into Memory Unit
 - First cycle is the Address Calculation
 - Other instructions need to wait: structural hazard

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
DEF10	9ABC	234 5678	1			

Excel Simulation – Cycle 4 (Execution)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

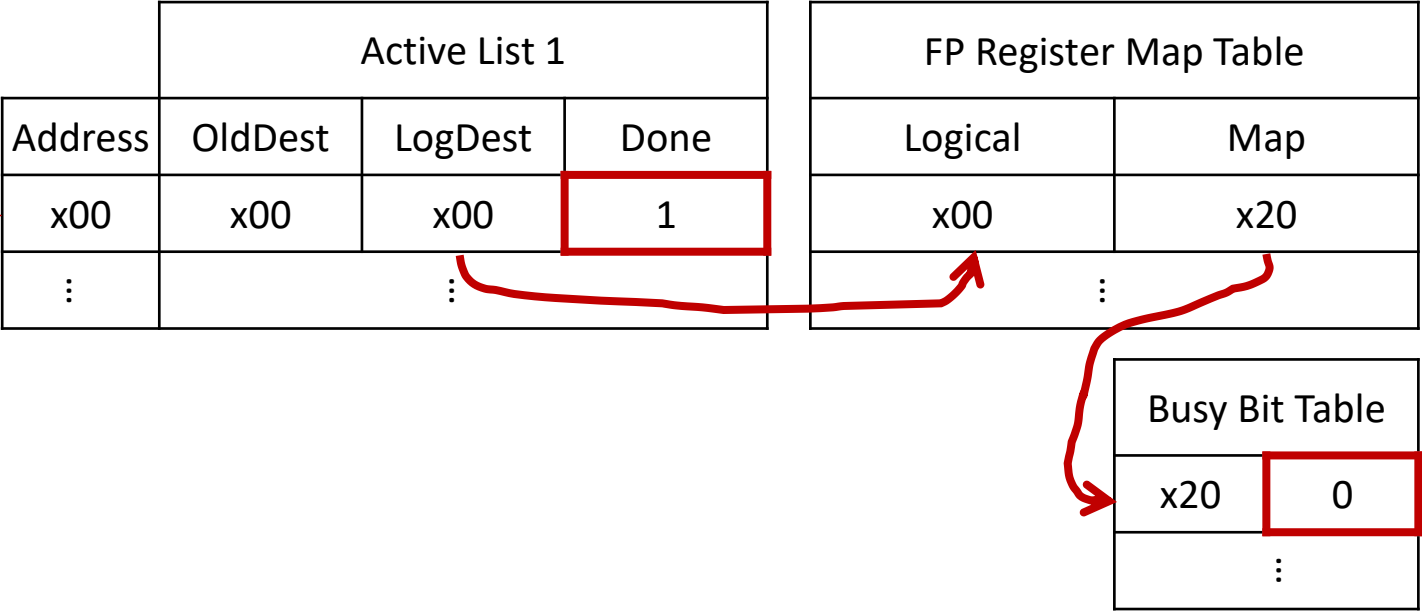
1. Instruction x01
 - Second cycle is the TLB - Load
2. Instruction x02
 - First cycle is the Address Calculation

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
	D E F 10	3 4 5 6 7 8 9 A B C	2	1		

Excel Simulation – Cycle 5 (Graduate)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4
#00000008	ADD.S	\$F5,	\$F0
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Get TAG from
Memory Unit

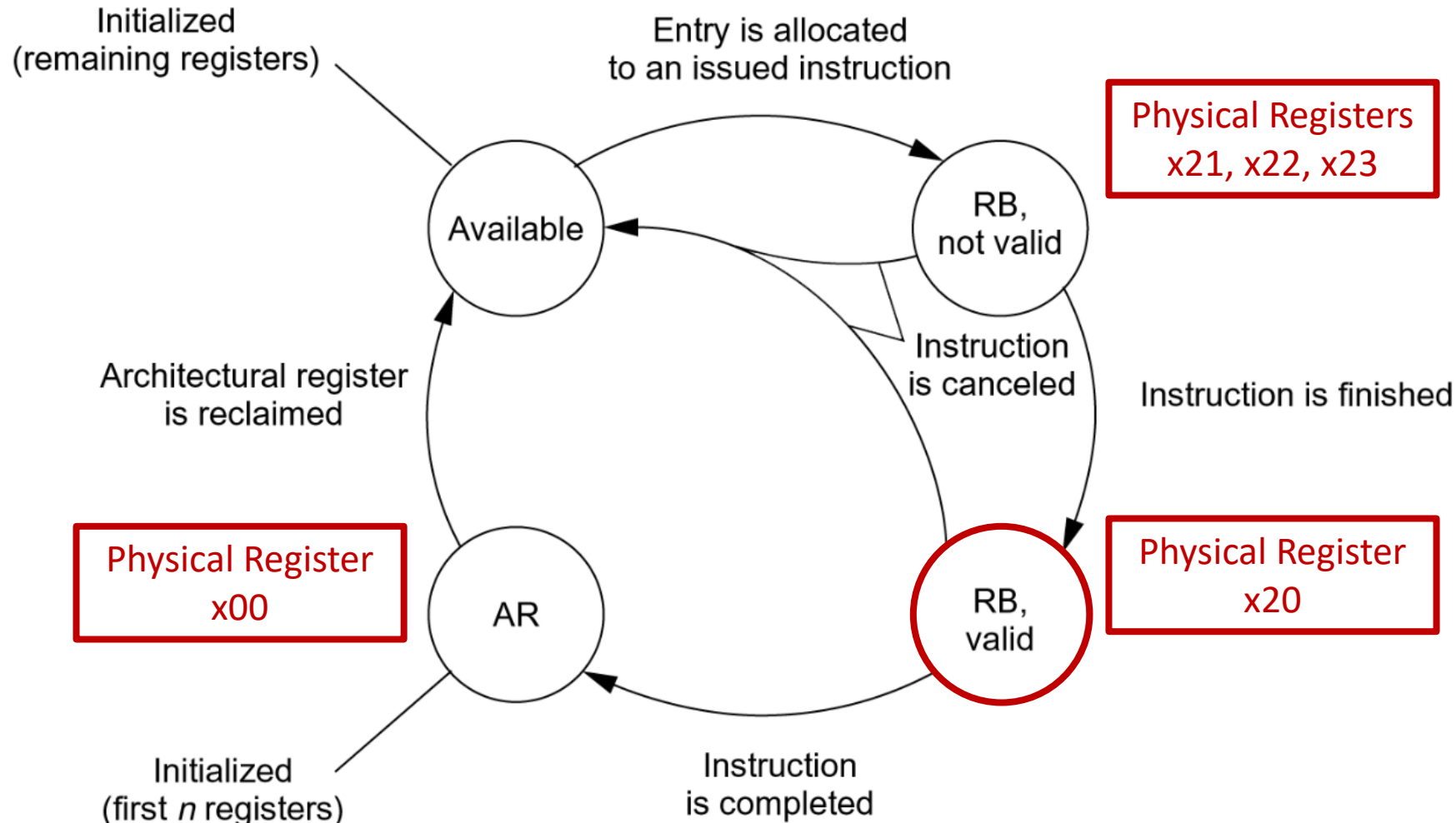


Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		4 5678 9ABC DEF10	3	2	1	

Instruction x01

- Active List: mark “Done” and get “LogDest”
 - Get physical register from Register Map Table and set busy to 0

Excel Simulation – Cycle 5 (Graduate)

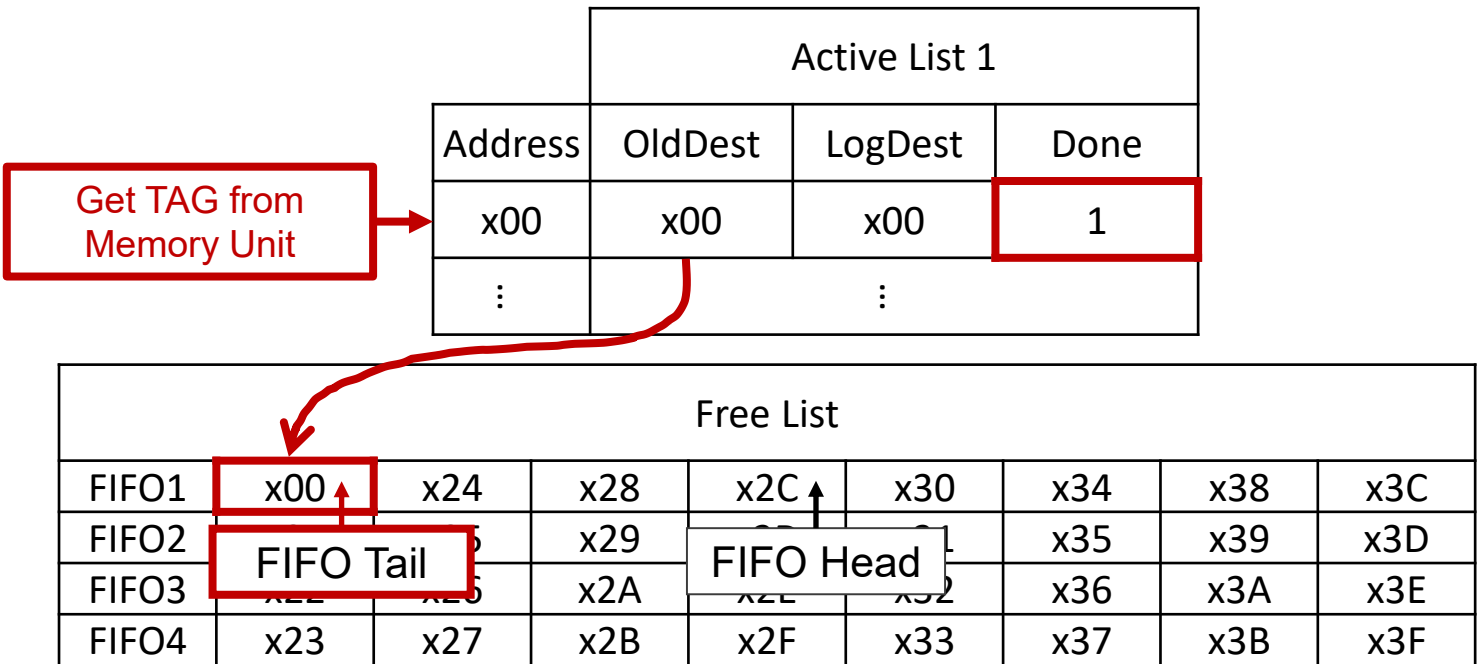


Busy Bit Table	
x20	0
⋮	

1 FSM per physical register

Excel Simulation – Cycle 5 (Graduate)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

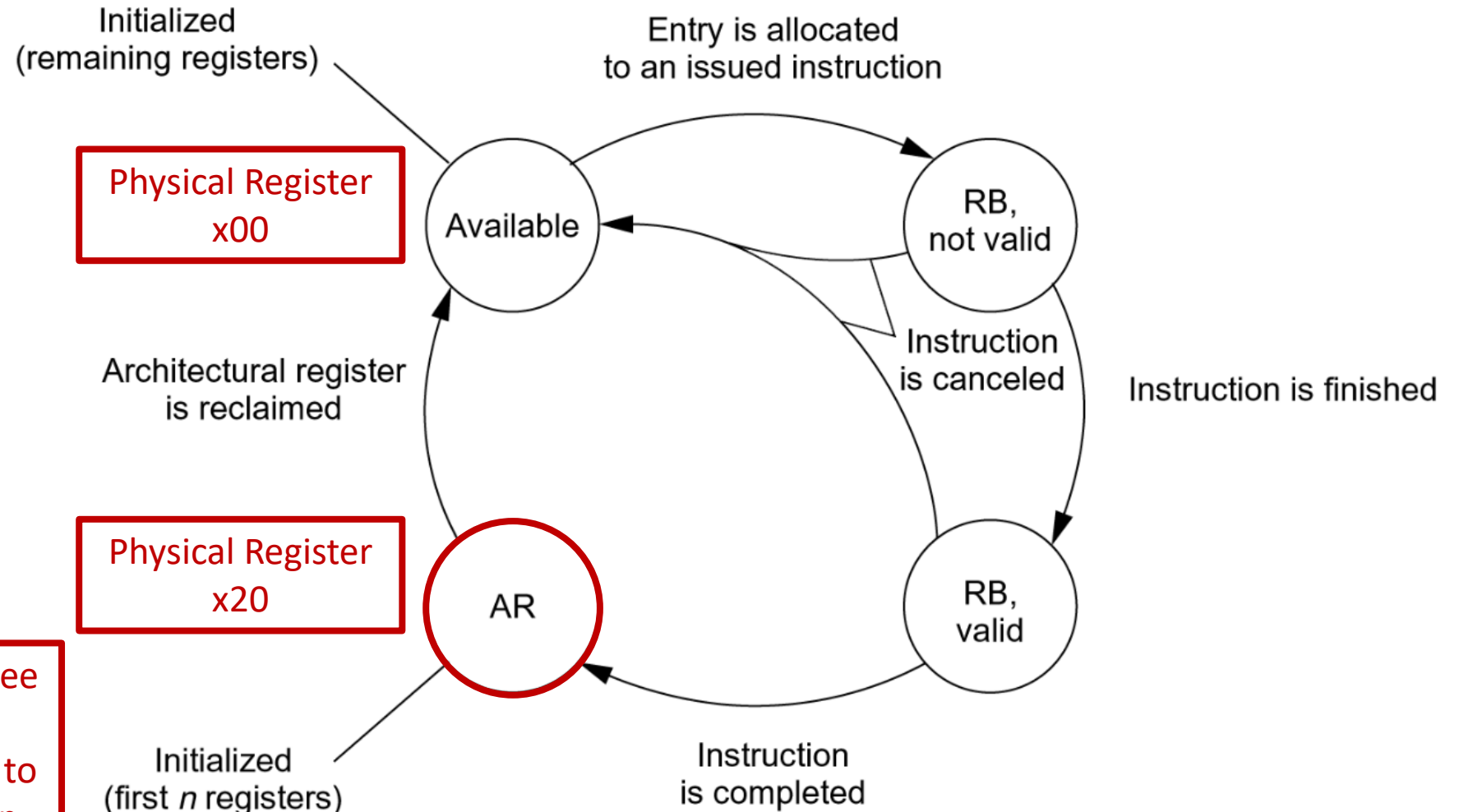


Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		4 5 6 7 8 9 A B C D E F 10	3	2	1	

Instruction x01

- Active List: if next to graduate, get “OldDest”
 - Move Free List Tail and set it to “OldDest”

Excel Simulation – Cycle 5 (Graduate)



OldDest (x00) is pushed to the Free List and x20 takes its place.

- No coming back: not possible to restore the original mapping in case of an exception or a branch misprediction

1 FSM per physical register

Excel Simulation – Cycle 6 (Graduate)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Active List 1			
Address	OldDest	LogDest	Done
x00			
⋮	⋮		

Address Queue			
Tag	Rdy	OpA	Dest
x08	1	I10x01	x21
x10	1	I10x01	x22
x18	1	I10x01	x23
⋮	⋮		

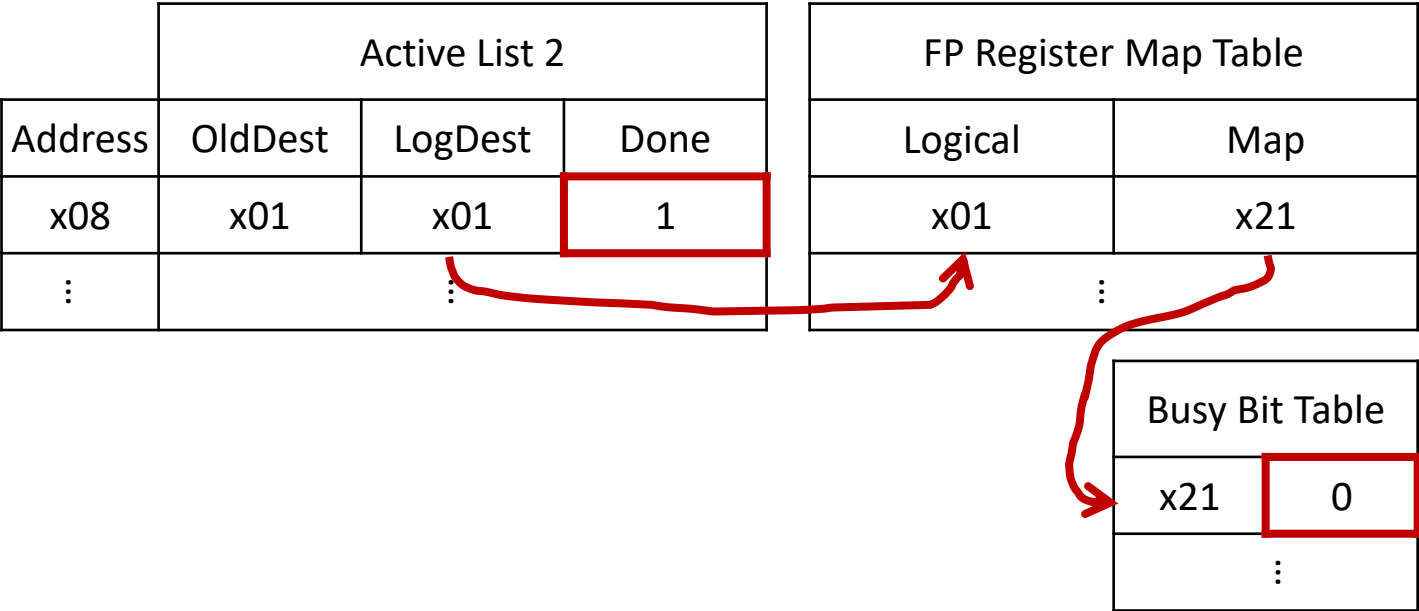
Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		5 6 7 8 9 ABC DEF 10	4	3	2	

1. Instruction x01
 - Remove entries from Address Queue and Active List

Excel Simulation – Cycle 6 (Graduate)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	
#00000009	MUL.S	\$F9,	
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

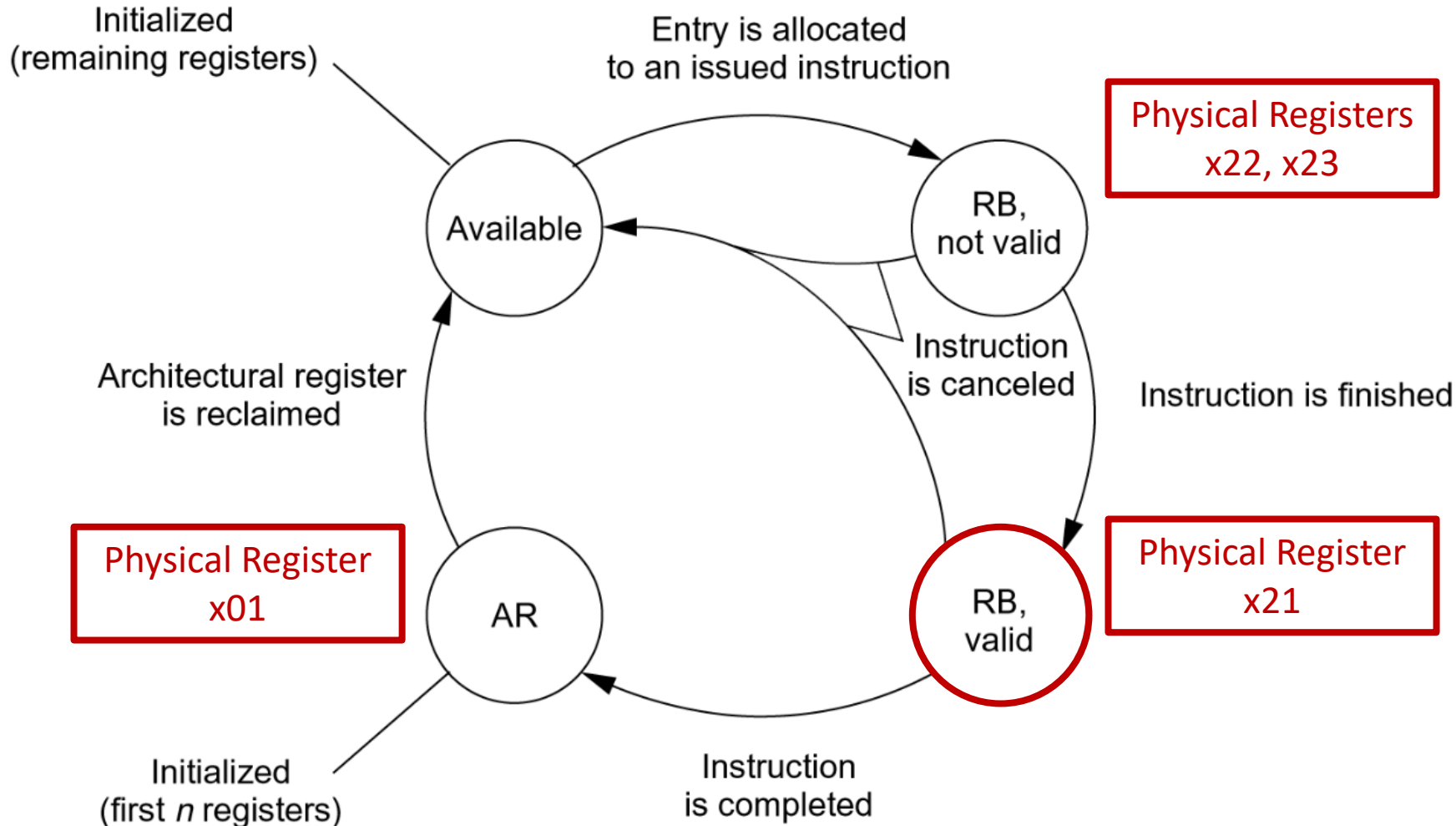
Get TAG from
Memory Unit



Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		5 6 7 8 9 ABC DEF 10	4	3	2	

2. Instruction x02
- Repeat same process as with x01
 - Instruction is Done and next to graduate

Excel Simulation – Cycle 6 (Graduate)



Busy Bit Table	
x20	0
⋮	

1 FSM per physical register

Excel Simulation – Cycle 6 (Graduate)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F0, \$F3
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Get TAG from Memory Unit

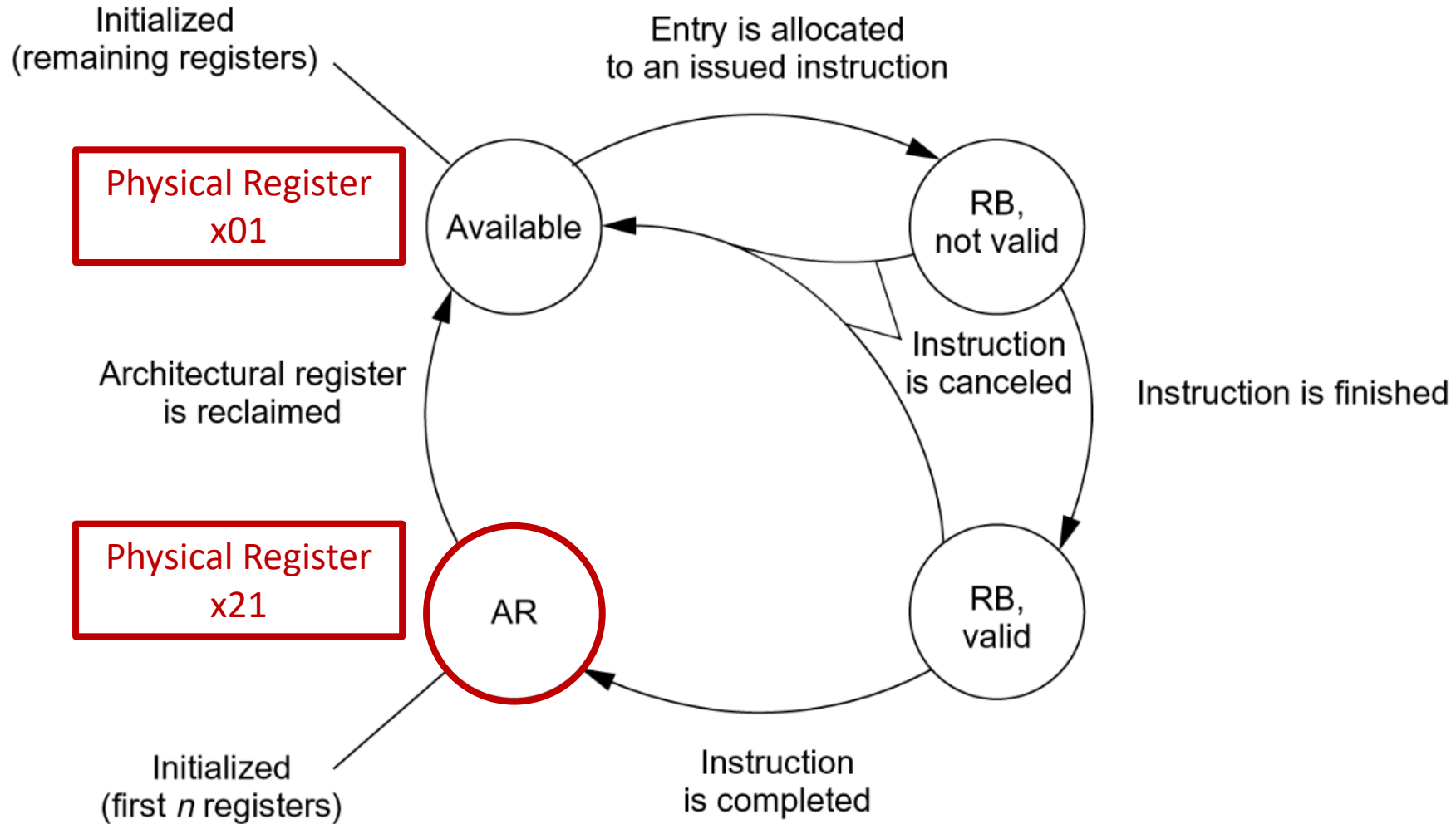
Active List 2			
Address	OldDest	LogDest	Done
x08	x01	x01	1
:		:	

Free List								
FIFO1	x00	x24	x28	x2C	x30	x34	x38	x3C
FIFO2	x01	x25	x29			x35	x39	x3D
FIFO3			x2A			x36	x3A	x3E
FIFO4			x2B	x2F	x33	x37	x3B	x3F

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		5 6 7 8 9 ABC DEF 10	4	3	2	

2. Instruction x02
- Repeat same process as with x01
 - Instruction is Done and next to graduate

Excel Simulation – Cycle 6 (Graduate)



1 FSM per physical register

Excel Simulation – Cycle 7 (Graduate)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Active List 2			
Address	OldDest	LogDest	Done
x08			
⋮	⋮		

Address Queue			
Tag	Rdy	OpA	Dest
x10	1	I10x01	x22
x18	1	I10x01	x23
⋮	⋮		

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		5 6 7 8 9 ABC DEF 10	8	4	3	

1. Instruction x02
 - Remove entries from Address Queue and Active List

Excel Simulation – Cycle 1 (Fetch)

```
#00000001  LDC1    $F0,    #0000($I1)
#00000002  LDC1    $F1,    #0001($I1)
#00000003  LDC1    $F2,    #0002($I1)
#00000004  LDC1    $F3,    #0003($I1)
#00000005  MUL.S   $F4,    $F0, $F3
#00000006  MUL.S   $F5,    $F1, $F2
#00000007  SUB.S   $F4,    $F4, $F5
#00000008  ADD.S   $F5,    $F0, $F1
#00000009  MUL.S   $F9,    $F5, $F5
#0000000A  MUL.S   $F10,   $F4, $F0
#0000000B  SUB.S   $F9,    $F9, $F10
#0000000C  SQRT.S  $F9,    $F9
#0000000D  SDC1    $F4,    #0004($I1)
#0000000E  SDC1    $F5,    #0005($I1)
#0000000F  SDC1    $F9,    #0006($I1)
#00000010  SDC1    $F10,   #0007($I1)
```

Back in time:

Focus on instructions x05 to x08

- RAW/WAW/WAR dependencies

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
5 6 7 8	1 2 3 4					

1. Fetch 4 instructions

Excel Simulation – Cycle 2 (Decode)

```
#00000001 LDC1    $F0,    #0000($I1)
#00000002 LDC1    $F1,    #0001($I1)
#00000003 LDC1    $F2,    #0002($I1)
#00000004 LDC1    $F3,    #0003($I1)
#00000005 MUL.S    $F4,    $F0, $F3
#00000006 MUL.S    $F5,    $F1, $F2
#00000007 SUB.S    $F4,    $F4, $F5
#00000008 ADD.S    $F5,    $F0, $F1
#00000009 MUL.S    $F9,    $F5, $F5
#0000000A MUL.S    $F10,   $F4, $F0
#0000000B SUB.S    $F9,    $F9, $F10
#0000000C SQRT.S   $F9,    $F9
#0000000D SDC1    $F4,    #0004($I1)
#0000000E SDC1    $F5,    #0005($I1)
#0000000F SDC1    $F9,    #0006($I1)
#00000010 SDC1    $F10,   #0007($I1)
```

Free List								
FIFO1	x20	x24	x28	x2C	x30	x34	x38	x3C
FIFO2	x21	x25	x29	x2D	x31	x35	x39	x3D
FIFO3	x22	x26	x2A	x2E	x32	x36	x3A	x3E
FIFO4	x23	x27	x2B	x2F	x33	x37	x3B	x3F

FIFO Tail

FP Register Map Table	
Logical	Map
⋮	⋮
x04	x26
x05	x27
⋮	⋮

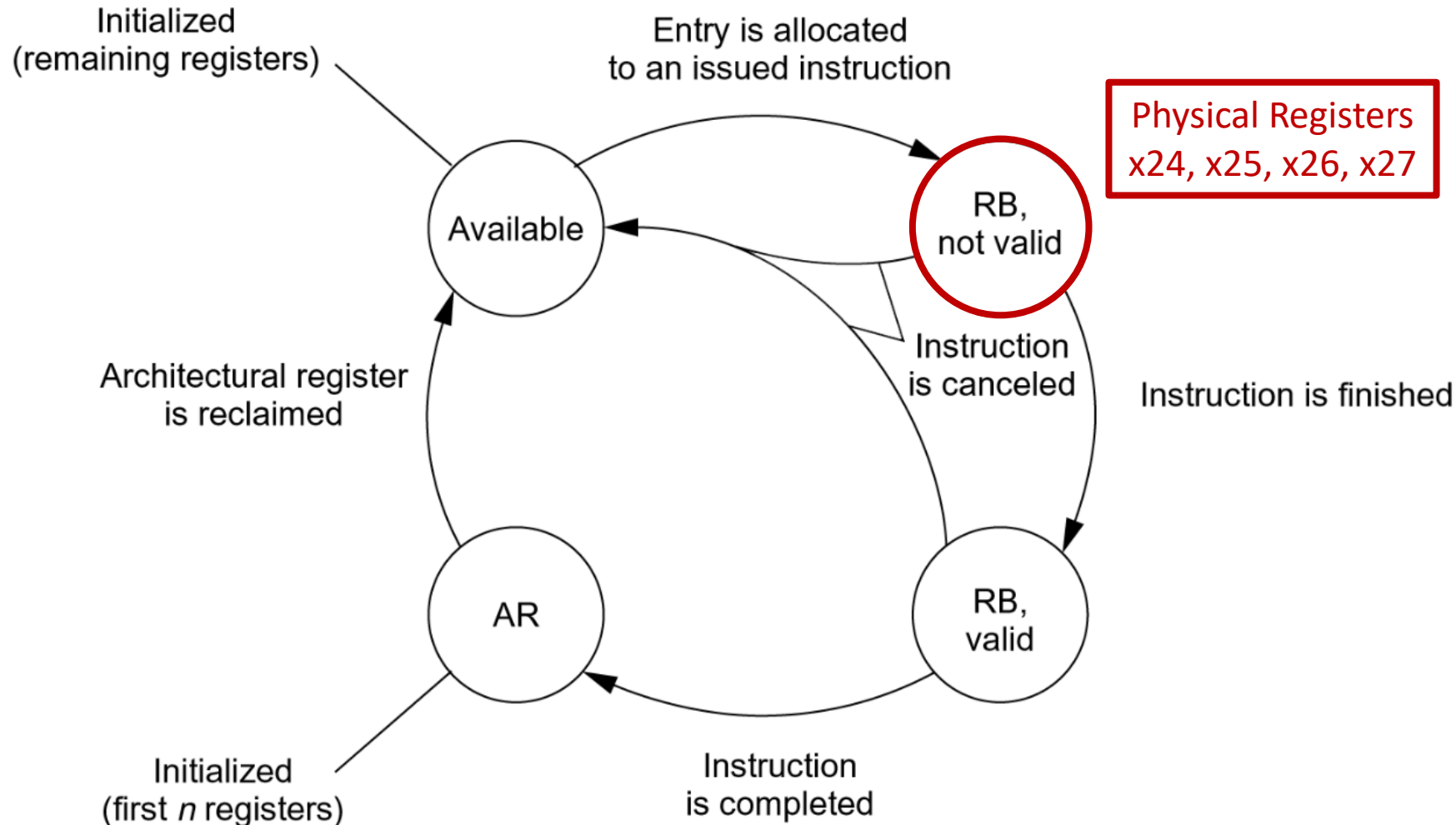
Only map registers
to latest value

Busy Bit Table	
⋮	
x24	1
x25	1
x26	1
x27	1
⋮	

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
9 A B C	5 6 7 8	1 2 3 4				

- WAW and WAR: \$F4 and \$F5 are used multiple times
 - Free 4 physical registers from the Free List and mark them as busy in the Busy Bit Table
 - Map registers in the Register Map Table

Excel Simulation – Cycle 2 (Decode)



Physical Registers
x24, x25, x26, x27

Busy Bit Table	
⋮	
x24	1
x25	1
x26	1
x27	1
⋮	

1 FSM per physical register

Excel Simulation – Cycle 2 (Decode)

```
#00000001 LDC1    $F0,    #0000($I1)
#00000002 LDC1    $F1,    #0001($I1)
#00000003 LDC1    $F2,    #0002($I1)
#00000004 LDC1    $F3,    #0003($I1)
#00000005 MUL.S   $F4,    $F0, $F3
#00000006 MUL.S   $F5,    $F1, $F2
#00000007 SUB.S   $F4,    $F4, $F5
#00000008 ADD.S   $F5,    $F0, $F1
#00000009 MUL.S   $F9,    $F5, $F5
#0000000A MUL.S   $F10,   $F4, $F0
#0000000B SUB.S   $F9,    $F9, $F10
#0000000C SQRT.S  $F9,    $F9
#0000000D SDC1    $F4,    #0004($I1)
#0000000E SDC1    $F5,    #0005($I1)
#0000000F SDC1    $F9,    #0006($I1)
#00000010 SDC1    $F10,   #0007($I1)
```

Active List 1			
Address	OldDest	LogDest	Done
⋮	⋮		
x01	x04	x04	0
⋮	⋮		

Active List 3			
Address	OldDest	LogDest	Done
⋮	⋮		
x11	x24	x04	0
⋮	⋮		

Active List 2			
Address	OldDest	LogDest	Done
⋮	⋮		
x09	x05	x05	0
⋮	⋮		

Active List 4			
Address	OldDest	LogDest	Done
⋮	⋮		
x19	x25	x05	0
⋮	⋮		

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
9 A B C	5 6 7 8	1 2 3 4				

2. Allocate entries in the Active Lists

- “OldDest” of instructions x07/x08 map to physical registers of instructions x05/x06

Excel Simulation – Cycle 3 (Issue)

```
#00000001 LDC1    $F0,    #0000($I1)
#00000002 LDC1    $F1,    #0001($I1)
#00000003 LDC1    $F2,    #0002($I1)
#00000004 LDC1    $F3,    #0003($I1)
#00000005 MUL.S   $F4,    $F0, $F3
#00000006 MUL.S   $F5,    $F1, $F2
#00000007 SUB.S   $F4,    $F4, $F5
#00000008 ADD.S   $F5,    $F0, $F1
#00000009 MUL.S   $F9,    $F5, $F5
#0000000A MUL.S   $F10,   $F4, $F0
#0000000B SUB.S   $F9,    $F9, $F10
#0000000C SQRT.S  $F9,    $F9
#0000000D SDC1    $F4,    #0004($I1)
#0000000E SDC1    $F5,    #0005($I1)
#0000000F SDC1    $F9,    #0006($I1)
#00000010 SDC1    $F10,   #0007($I1)
```

FP Queue					
Tag	Rdy	OpA	OpB	OpC	Dest
x01	x1 (b001)	x20	x23	0	x24
x09	x1 (b001)	x21	x22	0	x25
x11	x1 (b001)	x24	x25	0	x26
x19	x1 (b001)	x20	x21	0	x27
⋮					

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
DEF10	9ABC	5678	1234			

1. Push instructions to the FP Queue
 - Tag: Reference to Active List
 - Rdy: Are the operands ready? (OpC undefined => ready)
2. Read operands (OpA, OpB) from Register File
3. Remain in queue until all operands ready

Excel Simulation – Cycle 5 (Issue)

```
#00000001 LDC1    $F0,    #0000($I1)
#00000002 LDC1    $F1,    #0001($I1)
#00000003 LDC1    $F2,    #0002($I1)
#00000004 LDC1    $F3,    #0003($I1)
#00000005 MUL.S   $F4,    $F0, $F3
#00000006 MUL.S   $F5,    $F1, $F2
#00000007 SUB.S   $F4,    $F4, $F5
#00000008 ADD.S   $F5,    $F0, $F1
#00000009 MUL.S   $F9,    $F5, $F5
#0000000A MUL.S   $F10,   $F4, $F0
#0000000B SUB.S   $F9,    $F9, $F10
#0000000C SQRT.S  $F9,    $F9
#0000000D SDC1    $F4,    #0004($I1)
#0000000E SDC1    $F5,    #0005($I1)
#0000000F SDC1    $F9,    #0006($I1)
#00000010 SDC1    $F10,   #0007($I1)
```

Busy Bit Table					
x20	0	x22	1	x24	1
x21	1	x23	1	x25	1
⋮		⋮		⋮	

FP Queue					
Tag	Rdy	OpA	OpB	OpC	Dest
x01	x5 (b101)	x20	x23	0	x24
x09	x1 (b001)	x21	x22	0	x25
x11	x1 (b001)	x24	x25	0	x26
x19	x5 (b101)	x20	x21	0	x27
⋮					

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		4 5678 9ABC DEF10	3	2	1	

1. Instruction x01 finishes executing

- Operand \$F0 (Logical: x00 -> Physical: x20) is ready
- Mark as ready in FP Queue

Excel Simulation – Cycle 6 (Issue)

```
#00000001 LDC1 $F0, #0000($I1)
#00000002 LDC1 $F1, #0001($I1)
#00000003 LDC1 $F2, #0002($I1)
#00000004 LDC1 $F3, #0003($I1)
#00000005 MUL.S $F4, $F0, $F3
#00000006 MUL.S $F5, $F1, $F2
#00000007 SUB.S $F4, $F4, $F5
#00000008 ADD.S $F5, $F0, $F1
#00000009 MUL.S $F9, $F5, $F5
#0000000A MUL.S $F10, $F4, $F0
#0000000B SUB.S $F9, $F9, $F10
#0000000C SQRT.S $F9, $F9
#0000000D SDC1 $F4, #0004($I1)
#0000000E SDC1 $F5, #0005($I1)
#0000000F SDC1 $F9, #0006($I1)
#00000010 SDC1 $F10, #0007($I1)
```

Busy Bit Table					
x20	0	x22	1	x24	1
x21	0	x23	1	x25	1
⋮		⋮		⋮	

FP Queue					
Tag	Rdy	OpA	OpB	OpC	Dest
x01	x5 (b101)	x20	x23	0	x24
x09	x5 (b101)	x21	x22	0	x25
x11	x1 (b001)	x24	x25	0	x26
x19	x7 (b111)	x20	x21	0	x27
⋮					

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		5 6 7 8 9 A B C D E F 10	4	3	2	

- Instruction x02 finishes executing
 - Operand \$F1 (Logical: x01 -> Physical: x21) is ready
 - Mark as ready in FP Queue
- Instruction x08 is ready to be executed

Excel Simulation – Cycle 7 (Execute)

#00000001	LDC1	\$F0,	#0000(\$I1)
#00000002	LDC1	\$F1,	#0001(\$I1)
#00000003	LDC1	\$F2,	#0002(\$I1)
#00000004	LDC1	\$F3,	#0003(\$I1)
#00000005	MUL.S	\$F4,	\$F0, \$F3
#00000006	MUL.S	\$F5,	\$F1, \$F2
#00000007	SUB.S	\$F4,	\$F4, \$F5
#00000008	ADD.S	\$F5,	\$F0, \$F1
#00000009	MUL.S	\$F9,	\$F5, \$F5
#0000000A	MUL.S	\$F10,	\$F4, \$F0
#0000000B	SUB.S	\$F9,	\$F9, \$F10
#0000000C	SQRT.S	\$F9,	\$F9
#0000000D	SDC1	\$F4,	#0004(\$I1)
#0000000E	SDC1	\$F5,	#0005(\$I1)
#0000000F	SDC1	\$F9,	#0006(\$I1)
#00000010	SDC1	\$F10,	#0007(\$I1)

Busy Bit Table					
x20	0	x22	0	x24	1
x21	0	x23	1	x25	1
⋮		⋮		⋮	

FP Queue					
Tag	Rdy	OpA	OpB	OpC	Dest
x01	x5 (b101)	x20	x23	0	x24
x09	x7 (b111)	x21	x22	0	x25
x11	x1 (b001)	x24	x25	0	x26
⋮					

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7
		5 6 7 9 A B C D E F 10	8	4	3	

1. Instruction x08 starts executing: 3 cycles
 - Can leave FP Queue directly (unlike with Address Queue)
2. Instruction x03 finishes executing
 - Operand \$F2 (Logical: x02 -> Physical: x22) is ready
 - Mark as ready in FP Queue

Excel Simulation

- You have now all tools needed to complete every cycle of the simulation!
- Questions?
 - Do you have any case you would like to discuss here?

Outline

- Overview of the hand simulation of execution on the R10000 processor
 - Detailed review of the first 8 instructions (Done)
- Announcement of Exercise Set 1 (Done)
- **Next session: Q&A Homework, Lab, Exercise Set**